Nexa TEAM ECOBYTE

SDC HACKATHON 2026 · HOSTED BY XMUM AI CLUB

# Phase 1: Preliminary Project Proposal

Nexa shows workers which of their tasks AI is already doing, which skills carry them forward, and a cited path to a future role in three possible futures. HR gets the same view for the whole workforce.

TEAM
EcoByte

PROJECT
Nexa

DOMAIN
Domain 1 · Aiversecity Future-Ready Workforce

SDG TRACK
SDG 4 · Target 4.4 (+5, 8, 9, 10)

LIVE PROTOTYPE
https://nexa-xxx.vercel.app/

FILE
SDC2026_EcoByte_Preliminary.pdf

# 1. Team and Project Overview

| | |
|---|---|
| **Team Name** | EcoByte |
| **Selected Domain** | Domain 1 — Aiversecity Future-Ready Workforce |
| **Selected SDG Track (1–17)** | SDG 4 Quality Education (Target 4.4); also SDG 5, 8, 9 and 10 |
| **Project / Solution Name** | Nexa |
| **Name (Leader)** | Eya Hia |
| **Email & Phone (Leader)** | eyahia1712@gmail.com · +60 11-2511 4225 |
| **University Affiliation(s)** | Eya Hia — Lincoln University College<br>Ng Wei Meng — Universiti Kebangsaan Malaysia (UKM)<br>Alya Maisarah binti Jaafad Sadiq — Universiti Teknologi Malaysia (UTM)<br>Mohamad Khairul Haiqal bin Mohamad Razali — Universiti Malaysia Kelantan (UMK) |
| **Number of Team Members** | 4 — Eya Hia (leader), Ng Wei Meng, Alya Maisarah binti Jaafad Sadiq, Mohamad Khairul Haiqal bin Mohamad Razali |

**Live prototype:** https://nexa-xxx.vercel.app/
**Code repository:** https://github.com/eyahia1712/Nexa (public at submission)

# 2. Executive Summary & Problem Statement

## 2.1 The Problem Context

> *Describe the specific, real-world challenge your team is tackling. Who is affected, and why does it matter? [Maximum 250 words]*

AI is changing Malaysian work one task at a time, and almost nobody gets a personal answer about what that means for them. 56% of Malaysian professionals already use AI at work, but only 26% have had formal training for it (Hays Asia Salary Guide 2025). MOSTI estimates that more than 600,000 workers will need reskilling within three to five years, and the World Economic Forum expects 39% of core skills to change by 2030.

Three groups are affected. Workers in exposed roles, such as customer-service, clerical and data-entry staff, read headlines about AI but cannot tell which of their own tasks are changing, which skills still matter, or what to learn next. Graduates and TVET learners pick courses without knowing which roles will still be growing. HR and L&D teams must decide who to reskill, into what, how long it will take and whether it is fair, yet most still plan with spreadsheets and guesswork.

Existing tools forecast job titles instead of tasks, give one confident prediction, and recommend courses that are out of date by the time they are published. The result is wasted training budgets, anxious workers and a widening gap between people who can adapt and people who cannot. This matters for SDG 4 (relevant skills for decent work) and for SDG 5, 8 and 10: if reskilling reaches only the already advantaged, AI widens existing inequalities.

231 / 250 words

# 2. Executive Summary & Problem Statement

### 2.2 The Proposed AI Solution

> *High-level overview of the application, how it applies AI, and the SDC Phase 1 workshop tools it leverages. [Maximum 300 words]*

Nexa is a live web application that turns one CV into a clear, cited next step, and gives HR the same view for a whole workforce.

A person uploads a CV (PDF or Word) or picks a synthetic sample; Nexa reads it into skills from our catalogue and the person confirms each one. The engine breaks the person's job into official O*NET tasks and checks every task against measured real-world AI use from the Anthropic Economic Index, showing the share of their task time likely to be done with AI by 2030 in three scenarios (slow, moderate, fast) instead of one prediction. A Skills Radar scores 51 skills by momentum from WEF 2025, live arXiv research growth and AI-substitution pressure. Nexa then ranks 21 cited future roles with a four-part score anyone can add up by hand, shows the exact skills gap, marks roles that stay on top in all three futures, and builds a week-by-week pathway of 56 link-checked free or HRD Corp courses. The Impact Compass lets SDG priorities re-rank the options, and live arXiv and OpenDOSM signals update the plan, with every number dated.

The organisation view adds headcount, an org tree, each person's career story, team skills, reskilling cohorts, internal mobility, a fairness check and an SDG report; HR can load its own CSV, which stays in the browser.

AI is used where it adds evidence, not where it would guess: measured AI-use data drives the forecasts, an embedding model on a RunPod GPU adds semantic skill closeness, and an optional GenAI API (Claude) only maps CV text to catalogue skill ids. From the SDC Phase 1 workshops we apply GenAI APIs, data architecture (versioned, reproducible datasets), cloud deployment on Vercel, GPU compute on RunPod and Replit for ideation.

293 / 300 words

# 3. Technical Architecture & Implementation Plan

## 3.1 Conceptual Tech Stack

> *Technologies, frameworks, APIs and databases for a functional, deployed prototype, with a system block diagram (Appendix A).*

| Layer | Technology |
|---|---|
| Frontend | Next.js 16 (App Router), React 19, Tailwind CSS 4, Motion — responsive, phone to laptop |
| Backend | Next.js route handlers in TypeScript (Node.js): /api/parse-cv, /api/arxiv, /api/labour, /api/semantic |
| AI engine | Deterministic scoring engine: task shift in 3 scenarios, Skill Momentum, role matching, pathway, organisation analytics, fairness check |
| AI models & APIs | RunPod Serverless GPU: BAAI/bge-small-en-v1.5 embeddings (semantic skill closeness) · optional Claude API (CV text → catalogue skill ids) |
| Data | Versioned JSON built from O*NET 31.0 and the Anthropic Economic Index; WEF 2025, TalentCorp 2024; curated 21 roles, 51 skills, 56 courses, 25 SDG targets; live arXiv and OpenDOSM signals with dated snapshots. No personal-data database. |
| Hosting & DevOps | Vercel (auto-deploy on every GitHub push) · GitHub · Replit (ideation, .replit run config) |
| Quality | Vitest engine tests, catalogue-integrity tests, TypeScript, ESLint, automated course-link check |

### Did the project use an official SDC sponsor-provided GenAI API or Cloud Platform?

☑ Yes ☐ No

- **Replit** (Replit Core and AI credits): ideation with Replit Agent, where six project ideas were compared, and drafts of the editable pitch deck. The repository includes a `.replit` file, so Nexa also runs on Replit.
- **RunPod** (GPU credit): a Serverless GPU endpoint runs the open-source embedding model BAAI/bge-small-en-v1.5 to measure how close a person's skills are, in meaning, to each role's skills. The app calls it from `/api/semantic`.

# 3. Technical Architecture & Implementation Plan

## 3.2 Design-Thinking & Deployment Strategy

> *How the team applied design thinking to make the solution technically viable, user-friendly and ready for deployment by Phase 2. [Maximum 200 words]*

**Empathise.** We started from the personas in the SDC brief, a customer-service worker facing AI and an HR team planning reskilling, and wrote down their questions: “Am I next?”, “What should I learn?”, “Who do we reskill, into what, and is it fair?”

**Define.** The core need is a personal answer people can trust, so we set rules: forecast tasks, not titles; never invent a role, course or number; always show the source and date; show uncertainty as three futures.

**Ideate.** We compared six ideas on Replit and chose one engine that serves both people and organisations.

**Prototype.** We built the live app on Next.js and Vercel with synthetic data only, and checked every screen on phone and laptop.

**Test.** Walking teammates through the persona flows, we cut text-heavy screens, added “Why?” panels and a checkable score, and automated tests for the engine and course links.

**Deployment.** Nexa is already live, redeploys on every push and runs without any paid key, so Phase 2 can focus on Malaysian data (MASCO, job postings) and user pilots.

175 / 200 words

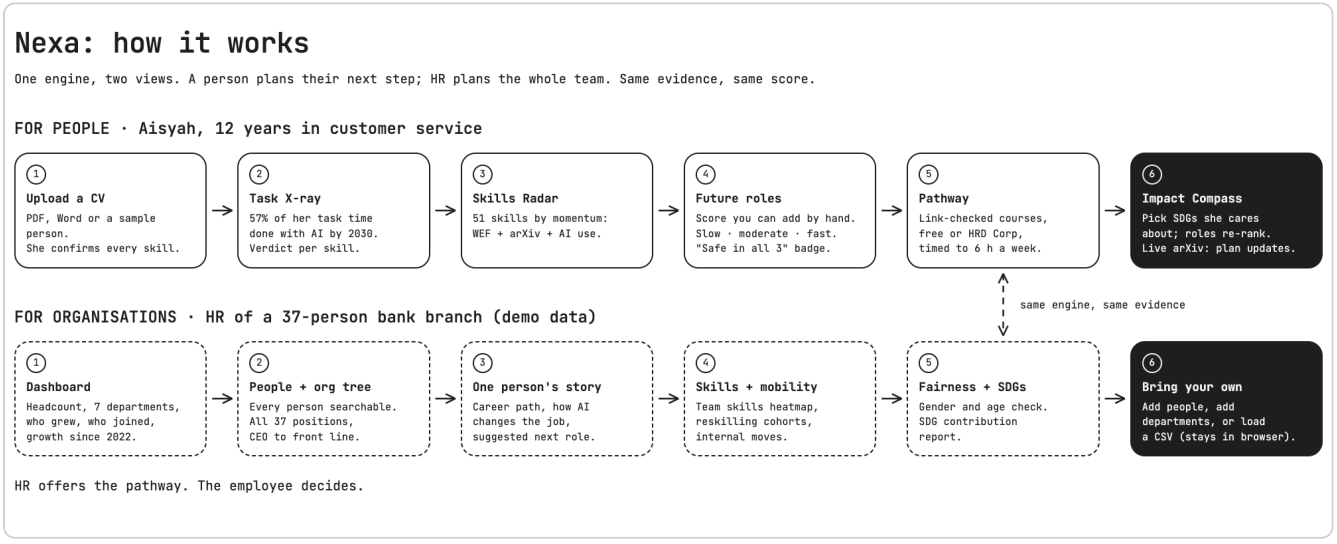



Figure 1. The two user journeys that came out of the empathise and define stages: one engine, a person's view and HR's view.

# 4. Preliminary Environmental & SDG Impact Framework

> *Impact baseline using published research or public benchmarks, with citations. Used by judges to assess Innovation & Creativity and Feasibility.*

| Metric Category | Our Response |
|---|---|
| **Target SDG Sub-Target(s)** | **Target 4.4:** "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship."<br><br>Supporting: **8.5** full and productive employment and decent work for all; **10.2** social and economic inclusion of all; **5.1** end discrimination against women (fairness check); **9.5** upgrade technological capabilities. |
| **Current Problem Baseline** | A **30-point AI training gap**: 56% of Malaysian professionals use AI at work, but only 26% are trained for it [1].<br>**600,000+** Malaysian workers need reskilling in the next three to five years [2].<br>**39%** of core skills are expected to change by 2030 [3].<br>Today these workers get **no personal, task-level forecast**, and HR reskilling plans are built in spreadsheets. |
| **Core Measurable KPI** | 1. **% of users** who finish one session with a named next role *and* a first verified course.<br>2. **Minutes** from CV upload to a cited, personalised plan.<br>3. **% of recommendations** with a checkable source, and the course-link pass rate. |
| **Phase 2 Impact Target** | In a pilot of at least 20 users (workers, graduates and one HR team):<br>• **≥ 80%** leave with a next role and a first course (against the 26% trained baseline);<br>• **≥ 90%** get a cited plan in **under 10 minutes**;<br>• **100%** of recommendations cite a source, and **≥ 95%** of course links pass the automated check. |

[1] Hays Asia Salary Guide 2025 (Malaysia). [2] Ministry of Science, Technology and Innovation (MOSTI), as cited in the SDC 2026 Domain 1 brief. [3] World Economic Forum, Future of Jobs Report 2025 (7 Jan 2025). Task data: O*NET 31.0 (U.S. DOL, CC BY 4.0); AI-use data: Anthropic Economic Index (2025-03-27 and 2026-03-05 releases, CC BY).

# 5. Eligibility & Readiness Checklist

> *Tick all boxes that apply before submitting.*

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study. *(Live: https://nexa-xxx.vercel.app/)*
- ☑ All fields in this template have been completed within the word limits stated.

# 6. IP & Originality Declaration

### 6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☑ Yes **☑ No**

Not applicable. Nexa is a new build: the repository, code, data pipeline, engine, interface and content were all created by Team EcoByte during the SDC 2026 hackathon period.

### 6.3 Employment-Related IP Conflicts

Is any team member bound by an employment, internship or consulting agreement under which their employer could claim ownership of work created during this hackathon?

☑ Yes **☑ No**

Not applicable. No team member is bound by such an agreement.

# 6. IP & Originality Declaration

### 6.2 Third-Party Code, Datasets, or Models

Does the project use third-party code libraries, open-source frameworks, datasets or pre-trained models that the team did not create?

☑ Yes ☐ No

| Name | Type (code / dataset / model) | License |
|---|---|---|
| Next.js, React, Tailwind CSS, Motion, Zod | Code | MIT |
| unpdf / mammoth | Code | MIT / BSD-2-Clause |
| Anthropic SDK, RunPod SDK | Code | MIT |
| sentence-transformers / PyTorch | Code | Apache-2.0 / BSD-3-Clause |
| Google Fonts: Archivo, Archivo Narrow, Geist Mono | Code (fonts) | SIL OFL 1.1 |
| BAAI/bge-small-en-v1.5 | Model | MIT |
| Claude (optional, via API) | Model | Anthropic Commercial Terms |
| O*NET 31.0 | Dataset | CC BY 4.0 |
| Anthropic Economic Index | Dataset | CC BY |
| arXiv metadata / OpenDOSM | Dataset | CC0 / CC BY 4.0 |
| WEF 2025, TalentCorp 2024, Hays 2025, MOSTI figures | Dataset (cited statistics) | Publisher terms, attributed |
| HRD Corp directory and public course pages (Coursera, edX, Kaggle Learn, Microsoft Learn and others) | Dataset (links only) | Each provider's terms, attributed |

All are declared and attributed in the repository README and on the app's `/evidence` page. Nexa itself is released under the Apache License 2.0.

# 7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

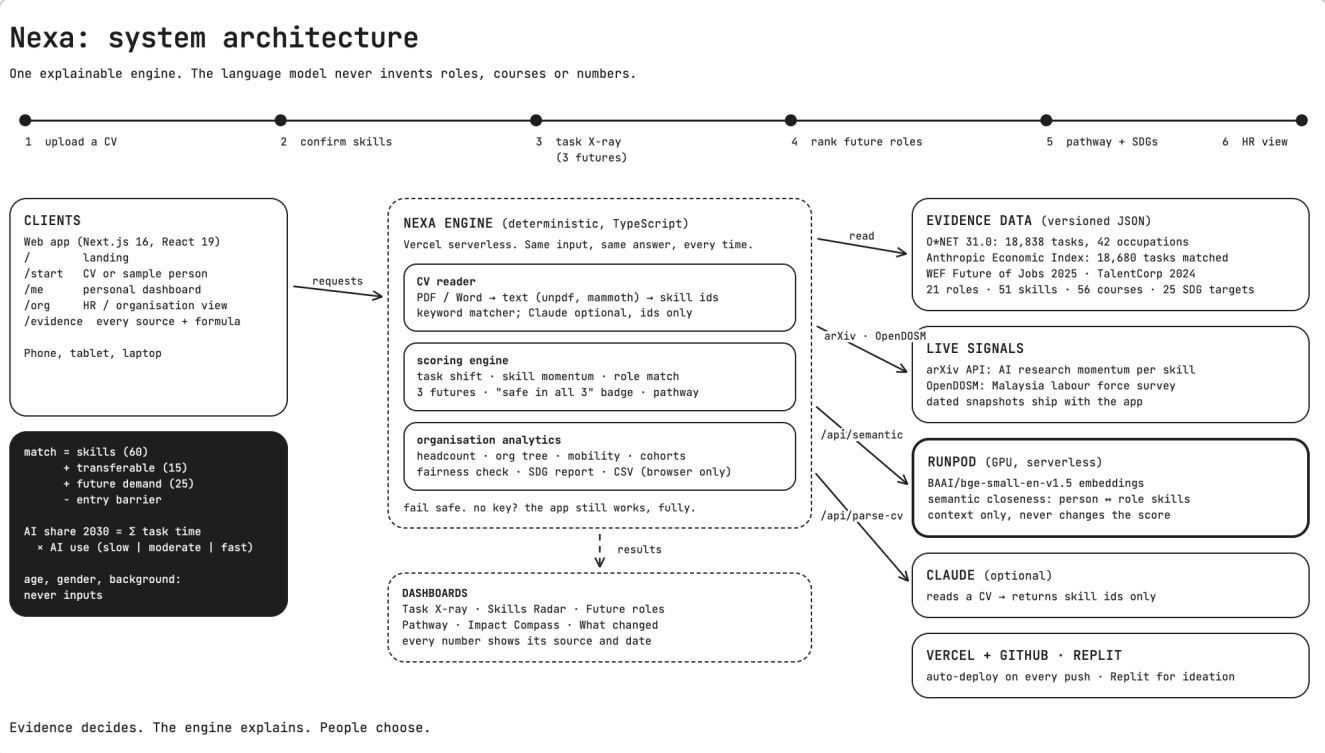
*Eya Hia*

**Eya Hia**, Team Leader, EcoByte

[Date]

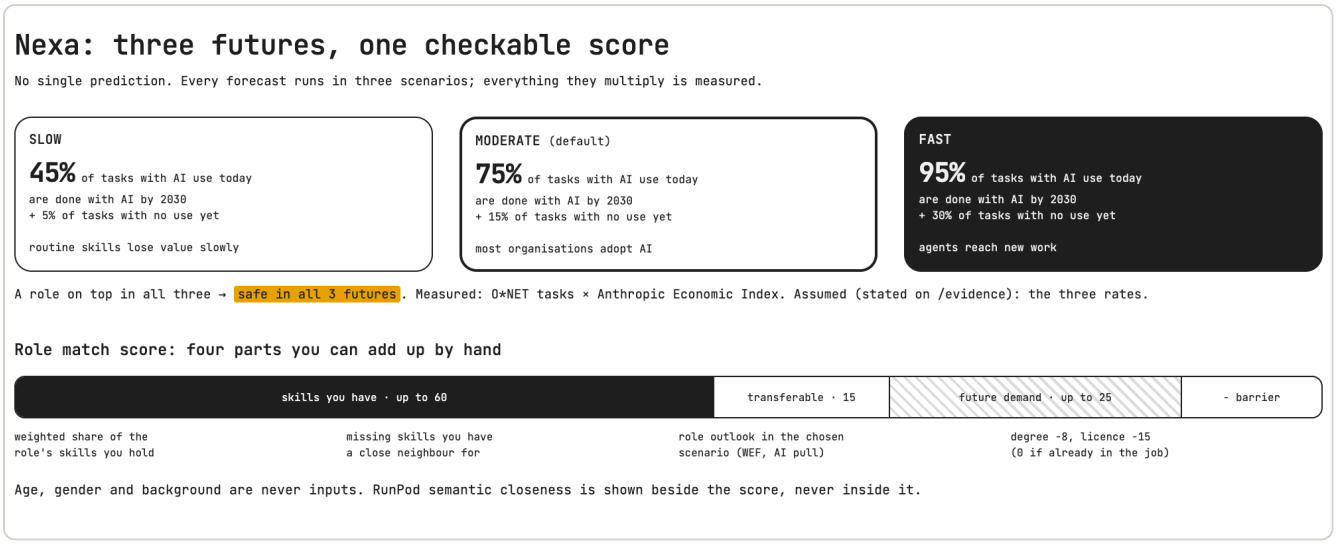
**24 October 2026**

# Appendix A · System Block Diagram




Clients → deterministic Nexa engine → evidence data, live signals, RunPod GPU and optional Claude; hosted on Vercel, auto-deployed from GitHub.

# Appendix B · Three futures, one checkable score




The scenario rates are stated assumptions; everything they multiply is measured. The role score has four visible parts that add up to the total.